Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Friendly Neighbourhood Devs
b. Team leader name: Keshav Lohiya
c. Problem Statement: Sustainability & Social Impact

Build an AI-powered solution that addresses a meaningful environmental or societal challenge. Solutions may focus on improving resource efficiency, increasing accessibility, strengthening resilience, supporting communities, or enabling more informed and sustainable decision-making.

Brief about the idea

LeafIt turns a hot city block into a costed, people-first cooling plan, built on real satellite data and planned with Gemini.

See the real place

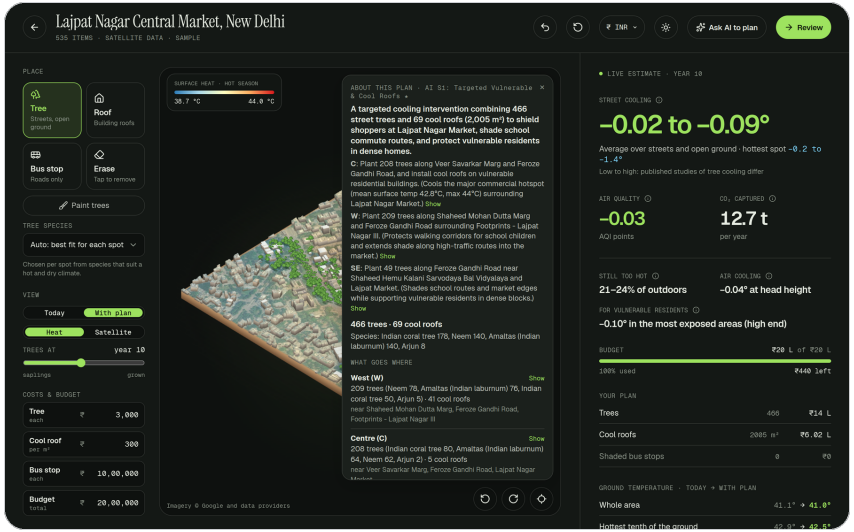
Mark any neighbourhood on Google's satellite map; LeafIt builds a 3D twin with its measured heat, trees, buildings and residents.

Plan in plain words

Place trees, cool roofs and shaded bus stops yourself, or give Gemini a brief and a budget and let it test the options.

Act on it

Before-and-after heat for each place, a bill of materials, a funding proposal and nearby vendors to quote.



Lajpat Nagar Central Market, New Delhi · Gemini's ₹20 lakh plan: 466 trees and 69 cool roofs

For **residents' associations, ward officers, NGOs and CSR teams** who fund cooling but can't test a plan before paying for it.

Opportunities

How it is different

Heat maps show where a city is hot. LeafIt shows what to do about it: which mix of trees, cool roofs and shelters, where, for whom and at what cost.

Starts from the shade already there, so trees are not wasted.

How it solves the problem

From one sentence and a budget to a tested, costed plan in about a minute, then on to a funding proposal and vendor quotes.

Guesswork becomes a plan you can defend.

What makes it unique

Honest numbers. Every coefficient is cited, cooling is shown as a low-to-high range, and every figure Gemini writes is checked against the simulator.

Gemini decides; the code calculates.

List of features offered by the solution

3D twin from real data

Satellite heat, existing trees, buildings, residents and roads, on 10-metre cells.

Design studio

Place trees, cool roofs and bus shelters; the numbers update instantly.

Gemini planner

Tests several plans, checks its own work and explains each zone.

People first

Cooling where young children and older residents live, not just the average.

Before and after

Measured heat today against the plan, place by place, in degrees.

Fund and build

Bill of materials, funding proposal and real local vendors with a ready quote.

Process flow diagram

USER JOURNEY

1 · Find
Search a place

2 · Mark
Drag a box

3 · Twin
Real data in 3D

4 · Plan
By hand or with Gemini

5 · Review
Before and after

6 · Act
Fund and build

ONE REAL RUN · LAJPAT NAGAR, NEW DELHI · ₹20 LAKH

The brief

"Keep shoppers and school children cool, and protect elderly residents in dense homes."

Gemini works

Reads the heat, trees and people; tests 3 plans; checks 2 of them itself.

The plan

466 trees and 69 cool roofs; 497 residents get a cooler roof.

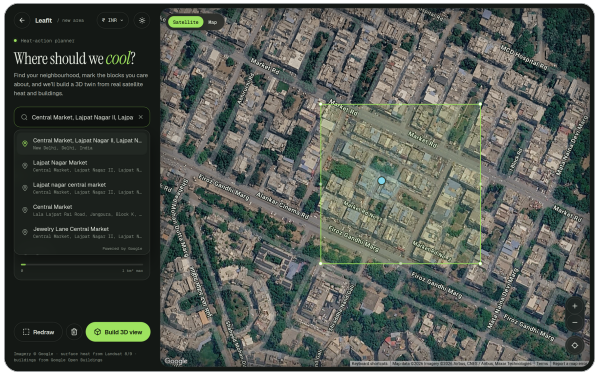
The result

Lajpat Nagar Market
42.7° → 41.9°
School 41.6° → 41.2°
Ground temperature, today → with plan

Ready to act

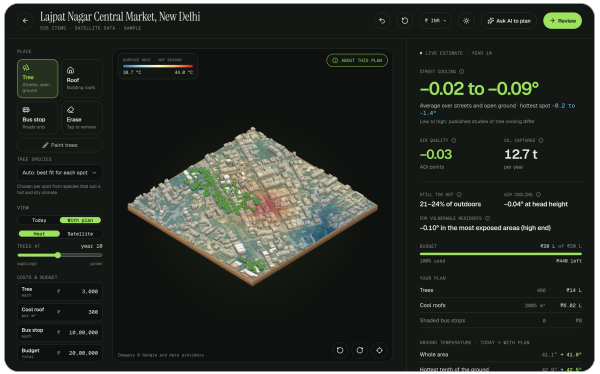
Bill of materials, funding proposal and local nurseries and contractors.

Wireframes and mock diagrams of the proposed solution



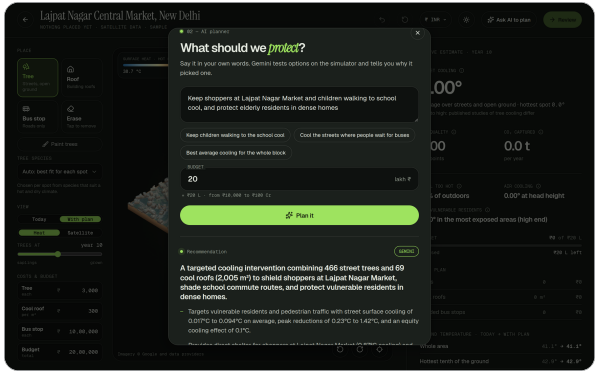
1 · Find and mark

Search a place and drag a box on the satellite map.



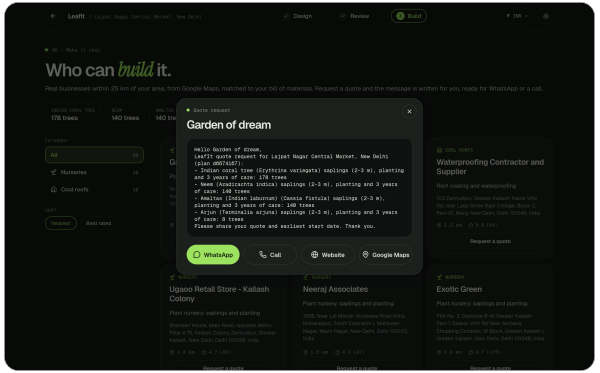
2 · Design in 3D

The area's real heat; place trees, roofs and shelters.



3 · Ask Gemini

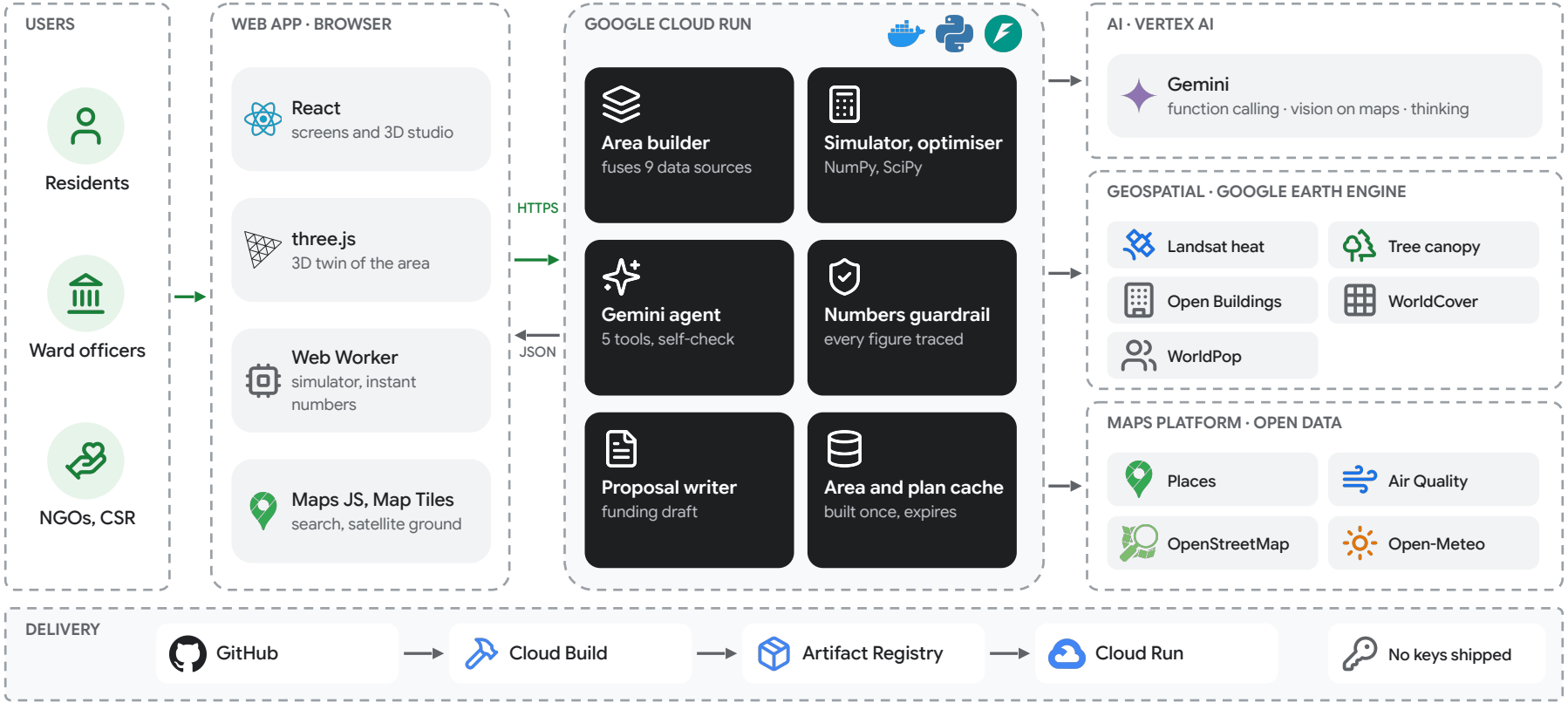
A brief and a budget in; a tested, explained plan out.



4 · Fund and build

Real local vendors and a quote message, ready to send.

Architecture diagram of the proposed solution



Technologies used in the solution

GOOGLE AI

- Gemini
- Vertex AI

GOOGLE CLOUD

- Cloud Run
- Cloud Build
- Earth Engine

GOOGLE MAPS PLATFORM

- Maps JS
- Places
- Map Tiles
- Air Quality

WEB APP

- React
- three.js
- Vite

SERVER

- Python
- FastAPI
- Docker

OPEN DATA

- OpenStreetMap
- Open-Meteo
- WorldPop
- WorldCover

Estimated implementation cost

Build once

Prototype to city pilot

2 developers × 3 months	₹7–9 L
Urban-forestry and GIS advice, part-time	₹1 L
Design, testing, field visits	₹0.5–1 L

₹8.5–11 L one time

Run · pilot

One city, ~300 AI plans and ~100 new areas a month

Cloud Run (free tier covers most of it)	₹0–500
Gemini, about ₹2 per AI plan	₹600–1,500
Maps Platform, inside free monthly limits	₹0
Earth Engine, noncommercial civic use	₹0

₹0.6–2 K per month

Run · city scale

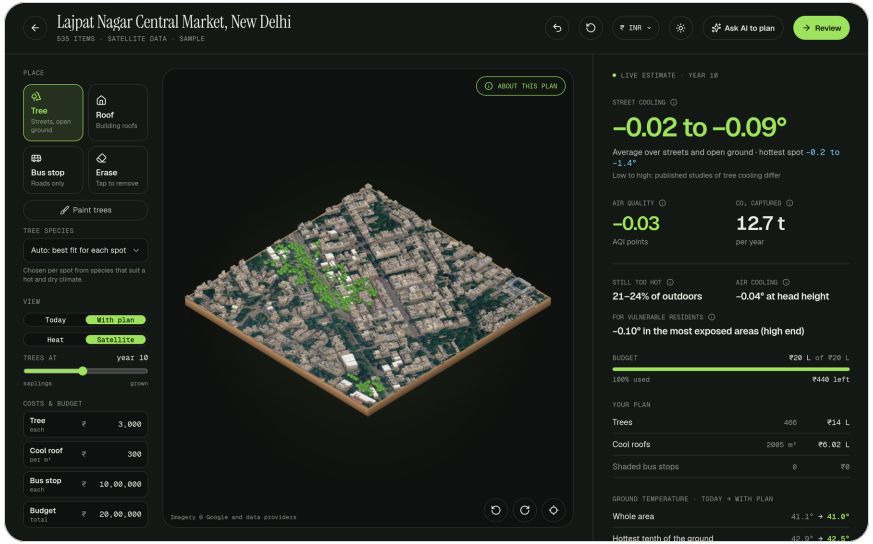
~5,000 AI plans and ~30,000 visits a month

Cloud Run	₹3–8 K
Gemini	₹10–25 K
Maps Platform, past the free limits	₹25–45 K
Earth Engine Basic plan ($500)	₹48 K

₹0.9–1.3 L per month

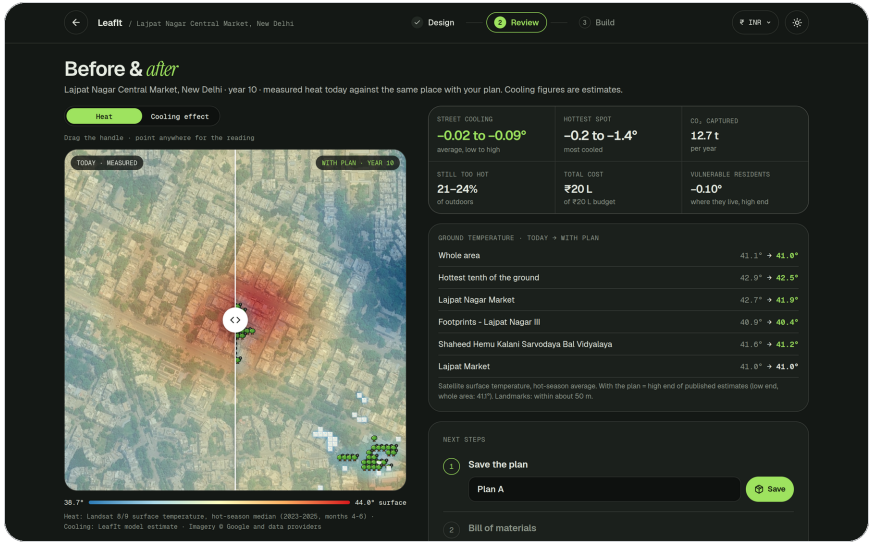
Estimates from Google's public price lists (Oct 2026) at ₹95 per US$. One AI plan measured at 8 Gemini calls, ~51,000 tokens in and ~2,200 out. Trees, roofs and shelters themselves are paid from each plan's own budget.

Snapshots of the prototype



Gemini's plan on the 3D twin

466 trees and 69 cool roofs on Lajpat Nagar's real satellite ground, with live numbers on the right.



Before and after

Measured heat today against the same place with the plan, and the temperature at each landmark.

Prototype performance and benchmarks

4 ms
to simulate a full plan
4,900 cells of 10 m

0.2 s
to optimise a plan
₹20 lakh of trees and roofs

~1 min
for Gemini to plan
3 plans tested, 8 calls

10–30 s
to build a new area
from live satellite data

158
automated tests pass
backend and web app

0
invented numbers
in Gemini's answer

GEMINI'S THREE OPTIONS · ₹20 LAKH

Option	Trees · roofs	Market	Under cool roofs
S1 ★	466 · 69	41.9°	497
S2	533 · 46	41.9°	368
S3	600 · 30	41.8°	not checked

★ Gemini's pick: the brief asked to protect elderly residents in dense homes, and it puts the most people under cool roofs. Market today: 42.7°.

Future development

Pilot with a Delhi ward

Plan, plant and measure one neighbourhood with a residents' association and the municipality.

Calibrate for Indian cities

Street sensors and afternoon thermal imagery to narrow today's cooling range.

Seasonal shade

Model leafless months for deciduous trees, using Indian growth data by species.

Track what gets planted

Record what is installed and its survival, then compare predicted and measured heat each summer.

More cities

Ready wherever Open Buildings has data: South and Southeast Asia, Africa and Latin America.

Mobile and local languages

A mobile app for field teams, with Hindi and regional languages.

Links

GitHub Public Repository

github.com/lohiyakeshav/LeafIt

Demo Video Link (3 Minutes)

drive.google.com/file/d/1mLXgTHEWch1o-BW0rdctq1ABphMi_sK8

Final Product Link

leafit.cloud.run

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

